Advanced Natural Language Processing (968G5): Practical Assignment (Max 5000 words) - Propaganda Detection

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Word count: XXXX

Introduction

Propaganda constitutes a biased representation of information which is intended to persuade others of a particular belief or worldview (Cambridge University Press & Assessment, n.d.). Whilst not necessarily false, propaganda risks engendering beliefs in individuals which do not adhere to matters of fact and that may serve to drive support for policies and societal attitudes which cause harm to indiviudals and groups, such as by reducing intent to vaccinate (Loomba et al., 2021). Social media provides a strong vector for propaganda to spread. As an example, this is particularly consequential when taking the form of state-sponsored "disinformation campaigns" (DiResta & Goldstein, 2025). Flagging propagandistic content to social media users may help to combat propagandistic content by encouraging users to be more critical of such content and reduce belief adoption based on misleading propaganda (Martel & Rand, 2023; Mena, 2026; but also see Grady et al., 2021). A reliance upon human-judges requires a set of impartial assessers to review (ideally) every social media post and flag its propagandistic nature. Such an approach is approximated by X in the form of community notes ("Community Notes," 2026), but whilst efficacy is generally good (Allen et al., 2024) coverage is far from ideal (Maldita.es, 2025), particularly during important political events (Center for Countering Digital Hate, 2024). Instead, automated detection may help to improve propaganda flagging.

INSERT A COUPLE OF EXAMPLES OF LITERATURE USING MNB/BERT FOR PROPAGANDA DETECTION

This project aims to address two tasks in propaganda detection. Firstly, two approahces to classifying known propagandistic texts as one of 8 propaganda variants will be evaluated, namely, a multinomial naive Bayes (MNB) model using a unigram-bigram bag-of-words representation, and a Bidirectional Encoder Representations from Transformers BERT (Devlin et al., 2019) model, fine-tuned to this project's propaganda text dataset.

Real-world applications of propaganda technique text classification likely will not evaluate text which is already known to be propagandistic. Thus, for the second task, we can augment the approach taken in the first task to include a preceding step to identify, if present, a propagandistic snippet within a given text. If such a snippet is identified, we can then classify the text with a specific propaganda label as in task 1, otherwise we label it as `not_propaganda`.

Methods

Task 1

- Unigram and Bigram BoW + Multinomial Naive Bayes
  - Laplace Smoothing (implemented by MNB function default `alpha=1`)
- BERT
- Metric: Accuracy
  - This is fine because class imbalance amongst propaganda labels is low

Approach A: Bag-of-Words with Multinomial Naive Bayes

$$P(c|d) = \frac{P(d|c)P(c)}{P(d)} \tag{1}$$

$$P(w_1, \dots, w_n|c) = P(w_1|c) \times \cdots \times P(w_n|c) \tag{2}$$

$$\hat{y} = \max_{c \in C} \left[\log P(c) + \sum_{i \in V} f_i \times \log \hat{P}(w_i|c) \right] \tag{3}$$

$$\hat{P}(w_i|c) = \frac{\text{freq}(w_i, c) + 1}{\sum_{w \in V} (\text{freq}(w, c) + 1)} \tag{4}$$

Approach B: Fine-tuned BERT

Task 2

Exploratory Data Analysis

In machine learning classification tasks, the distribution of classes in the training data can influence the manner in which models discriminate between classes. For instance, suppose a class in the training data equates to 95% of all labels present. A model which only predicts this single class will achieve an accuracy of 95%. Thus, it is appropriate to evaluate the class distribution before considering accuracy as an appropriate evaluation metric for text classification.

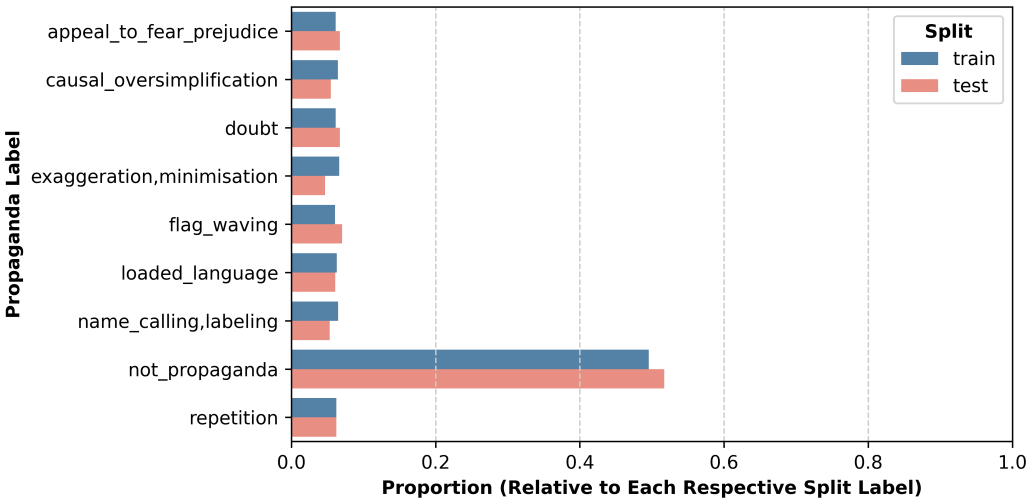


Figure 1. Comparison of Multiclass Distribution Between Train and Test Datasets

Label	Snippet	Context
appeal_to_fear_prejudice	church, one, would, the church, people, u, u s, could, us, god	said, many, letter, us, trump, plague, could, people, bishops, even
causal_oversimplification	one, god, cia, church, the cia, the church, trump, clearly, want, would	said, one, would, security, like, words, national, trump, national security, man
doubt	guardian, the guardian, people, even, evidence, know, trump, never, story, church	said, church, question, also, cannot, kavanaugh, people, viganò, pope, catholic
exaggeration,minimisation	world, one, history, ever, thing, the world, church, without, even, life	one, trump, said, president, people, one of, church, even, many, former
flag_waving	american, people, america, american people, the american, country, americans, protect, nation, u	trump, said, continue, president, public, put, mayors, iran, cia, safety
loaded_language	church, people, life, smear, whoa, scandal, pope, last, agenda, damn	trump, church, said, would, also, new, many, the church, one, people
name_calling,labeling	trump, leftist, left, america, disgrace, anti, infidels, partisan, priest, disgrace to	trump, one, would, church, news, assange, catholic, people, like, must
repetition	aliens, people, autonomy, trump, illegal, illegal aliens, muslim, invaders, never, become	one, said, parents, trump, many, obama, school, president, immigration, even

Table 1. Most Common Unigrams/Bigrams by Label

Table 1

Results

Task 1: Propaganda Technique Classification

Multinomial Naive Bayes Plain Text Accuracy: 0.41

Label	Most Discriminative Features
appeal_to_fear_prejudice	if, we, iran, face, will, letter, the easiest, senior labour, allies, die
causal_oversimplification	because, she, clearly, it is, the cia, cia, if, sin, he, it
doubt	guardian, the guardian, why, ?, but, evidence, he, t, harding, was
exaggeration,minimisation	the most, most, world s, thing, history, weapons to, denied, the worst, ever, franken
flag_waving	american people, the american, american, our, america, americans, mayors, continue to, continue, people
loaded_language	evidence, that no, it was, wills, region, house, ratzinger, handling of, staff, damn
name_calling,labeling	jews, account, nothing, account was, news and, allah, when he, as the, assange, 11
repetition	parents, aliens, school, curriculum, autonomy, parents are, document, hat, illegal aliens, teachers

Table 2. Most Discriminative Words by Label

BERT Accuracy: 0.51

Task 2: Propaganda Span and Technique Identification

Conclusion

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